

THREAD

Pemrograman berorientasi objek II

Ismael, S.Kom, M. Kom



CONTOH PROGRAM THREAD (MAIN THREAD)

```
public class DemoThread {  
    public static void main(String[] args) throws InterruptedException {  
        System.out.println("Nama thread utama : " + Thread.currentThread().getName());  
    }  
}
```

run:

Nama thread utama : main

BUILD SUCCESSFUL (total time: 0 seconds)



CONTOH PROGRAM THREAD (MAIN THREAD)

```
public class DemoThread {  
    public static void main(String[] args) throws InterruptedException{  
        String namaThreadUtama = Thread.currentThread().getName();  
        System.out.println("Nama thread utama : " + namaThreadUtama);  
  
        // melayani 10 pelanggan secara berurutan  
        for (int pelanggan = 1; pelanggan <= 10; pelanggan++) {  
            System.out.println(namaThreadUtama + " melayani pelanggan nomor " + pelanggan);  
            Thread.sleep(1000);  
        }  
    }  
}
```

```
.....  
Nama thread utama : main  
main melayani pelanggan nomor 1  
main melayani pelanggan nomor 2  
main melayani pelanggan nomor 3  
main melayani pelanggan nomor 4  
main melayani pelanggan nomor 5  
main melayani pelanggan nomor 6  
main melayani pelanggan nomor 7  
main melayani pelanggan nomor 8  
main melayani pelanggan nomor 9  
main melayani pelanggan nomor 10  
BUILD SUCCESSFUL (total time: 10 seconds)
```



CONTOH MULTI THREAD (RUNNABLE)

- Buat sebuah kelas dengan nama KasirSatu, modifikasi sehingga seperti berikut :

```
public class KasirSatu implements Runnable{
    @Override
    public void run() {
        String namaThreadBaru = Thread.currentThread().getName();
        System.out.println("Nama thread baru : " + namaThreadBaru);

        // kasir baru melayani pelanggan nomor 7 sampai 12
        for (int pelanggan = 7; pelanggan <= 12; pelanggan++) {
            System.out.println(namaThreadBaru + " melayani pelanggan nomor " + pelanggan);
        }
    }
}
```



CONTOH MULTI THREAD (LANJUTAN)

- Buat sebuah kelas dengan nama Kasir (Main Program), modifikasi sehingga seperti berikut :

```
public class Kasir {  
    public static void main(String[] args) throws InterruptedException {  
        String namaThreadUtama = Thread.currentThread().getName();  
        System.out.println("Nama thread utama : " + namaThreadUtama);  
  
        Thread kasirBaru = new Thread(new KasirSatu());  
        kasirBaru.start();  
  
        for (int pelanggan = 1; pelanggan <= 6; pelanggan++) {  
            System.out.println(namaThreadUtama + " melayani pelanggan nomor " + pelanggan);  
            Thread.sleep(1000);  
        }  
    }  
}
```



CONTOH MULTI THREAD

- Jalankan program kasir, perhatikan outputnya

```
run:
Nama thread utama : main
Nama thread kasir 1 : Thread-0
main melayani pelanggan nomor 1
Thread-0 melayani pelanggan nomor 7
Thread-0 melayani pelanggan nomor 8
main melayani pelanggan nomor 2
Thread-0 melayani pelanggan nomor 9
main melayani pelanggan nomor 3
Thread-0 melayani pelanggan nomor 10
main melayani pelanggan nomor 4
main melayani pelanggan nomor 5
Thread-0 melayani pelanggan nomor 11
Thread-0 melayani pelanggan nomor 12
main melayani pelanggan nomor 6
BUILD SUCCESSFUL (total time: 6 seconds)
```

```
run:
Nama thread utama : main
Nama thread kasir 1 : Thread-0
main melayani pelanggan nomor 1
Thread-0 melayani pelanggan nomor 7
Thread-0 melayani pelanggan nomor 8
main melayani pelanggan nomor 2
Thread-0 melayani pelanggan nomor 9
main melayani pelanggan nomor 3
main melayani pelanggan nomor 4
Thread-0 melayani pelanggan nomor 10
main melayani pelanggan nomor 5
Thread-0 melayani pelanggan nomor 11
main melayani pelanggan nomor 6
Thread-0 melayani pelanggan nomor 12
BUILD SUCCESSFUL (total time: 6 seconds)
```

```
run:
Nama thread utama : main
Nama thread kasir 1 : Thread-0
main melayani pelanggan nomor 1
Thread-0 melayani pelanggan nomor 7
Thread-0 melayani pelanggan nomor 8
main melayani pelanggan nomor 2
Thread-0 melayani pelanggan nomor 9
main melayani pelanggan nomor 3
main melayani pelanggan nomor 4
Thread-0 melayani pelanggan nomor 10
main melayani pelanggan nomor 5
Thread-0 melayani pelanggan nomor 11
Thread-0 melayani pelanggan nomor 12
main melayani pelanggan nomor 6
BUILD SUCCESSFUL (total time: 6 seconds)
```



CONTOH MULTI THREAD (EXTEND TRHEAD)

- Buat sebuah class baru dengan nama TestThread

```
class ThreadDemo extends Thread {
    private Thread t;
    private String threadName;

    ThreadDemo( String name) {
        threadName = name;
        System.out.println("Creating " + threadName );
    }

    public void run() {
        System.out.println("Running " + threadName );
        try {
            for(int i = 4; i > 0; i--) {
                System.out.println("Thread: " + threadName + ", " + i);
                // Let the thread sleep for a while.
                Thread.sleep( millis:50);
            }
        } catch (InterruptedException e) {
            System.out.println("Thread " + threadName + " interrupted.");
        }
        System.out.println("Thread " + threadName + " exiting.");
    }

    public void start () {
        System.out.println("Starting " + threadName );
        if (t == null) {
            t = new Thread ( task:this, name:threadName);
            t.start ();
        }
    }
}
```

```
public class TestThread {
    public static void main(String[] args) {
        ThreadDemo T1 = new ThreadDemo( name:"Thread-1");
        T1.start();

        ThreadDemo T2 = new ThreadDemo( name:"Thread-2");
        T2.start();
    }
}
```



CONTOH MULTI THREAD (EXTEND TRHEAD)

jalankan kode TestThread untuk melihat hasilnya

```
run:
Creating Thread-1
Starting Thread-1
Creating Thread-2
Starting Thread-2
Running Thread-1
Running Thread-2
Thread: Thread-2, 4
Thread: Thread-1, 4
Thread: Thread-2, 3
Thread: Thread-1, 3
Thread: Thread-2, 2
Thread: Thread-1, 2
Thread: Thread-2, 1
Thread: Thread-1, 1
Thread Thread-1 exiting.
Thread Thread-2 exiting.
BUILD SUCCESSFUL (total time: 0 seconds)
```

```
run:
Creating Thread-1
Starting Thread-1
Creating Thread-2
Starting Thread-2
Running Thread-1
Running Thread-2
Thread: Thread-1, 4
Thread: Thread-2, 4
Thread: Thread-1, 3
Thread: Thread-2, 3
Thread: Thread-2, 2
Thread: Thread-1, 2
Thread: Thread-2, 1
Thread: Thread-1, 1
Thread Thread-1 exiting.
Thread Thread-2 exiting.
BUILD SUCCESSFUL (total time: 0 seconds)
```

```
run:
Creating Thread-1
Starting Thread-1
Creating Thread-2
Starting Thread-2
Running Thread-1
Running Thread-2
Thread: Thread-2, 4
Thread: Thread-1, 4
Thread: Thread-1, 3
Thread: Thread-2, 3
Thread: Thread-2, 2
Thread: Thread-1, 2
Thread: Thread-1, 1
Thread: Thread-2, 1
Thread Thread-1 exiting.
Thread Thread-2 exiting.
BUILD SUCCESSFUL (total time: 0 seconds)
```



LATIHAN

- Coba tambahkan satu lagi thread baru dengan nama Kasirdua. Kemudian panggil kasirdua pada main program,
- Coba jalankan amati perbedaan output dari programnya.

